

Roadmap: the berries and their sequencing — rev 2

Rev 2 (2026-08-10): refreshed after B2 completed and B4 phases 1–2 landed; folds in the Shannon programme (aips/draft/shannon-programme.md) and the decisions of 2026-08-10 (slots 1–3, Euclidean variants, dirty-model direction).

1. The problem

The project has many individually exciting outputs but no ordering. This AIP names the target outputs (“berries”, session 2026-08-09), their dependency structure, and a sequence, so tracks can be cut against it instead of re-derived per session.

2. The berries — status

Landing dyadic-machine-v0 was the trunk; it is landed and mechanized.

- **B2 — the theorem trio. DONE** (2026-08-10): streaming, random access + optimality, locality composition, RAM simulations in *both* directions (target 3 closed). 32 receipts; Expo 1 carries it fully lean-backed. By-product decision still open: standalone machine-model paper (q2 below, now ripe).
- **B4 — Kraft/distance accounting. Phases 1–2 DONE** (kraft_iff_mounting both directions, distance zip, entropy achievability + Gibbs converse; Expo 2 “Heads at a distance”). Remaining: phase 3 cliff witness (needs k-way merge machinery + cache-v0 call 2), size-aware distances (v1), [72] — which may resolve in the dirty layer instead (cache-v0 §4b).
- **B4’ — the Shannon programme** (new, successor line to B4): the entropy pair is the source coding theorem; the programme is the *combinatorial half of information theory, mechanized* (shannon-programme §7). Concrete berries: move 1 (block-adaptive re-mounting = entropy rate — SLOTTED), move 2 (universal mounting), the C = 1 paragraph (free), Campbell/Rényi under level prices (with the Euclidean variants below).
- **Geometry variants** (decided 2026-08-10): after the C = 1 identity, do the Euclidean variants *immediately* — $c(\ell) = 2^{(\ell/2)}$ and $2^{(\ell/3)}$ (k = 2, 3) as first-class model variants next to $c \equiv 1$, not a distant v1 nicety. The level-price dial is a choice of ambient dimension (hyperbolic at $c \equiv 1$, k-Euclidean at $2^{(\ell/k)}$, fractal at non-integer exponent); folklore anchors in shannon-programme §2.
- **Dirty model** (direction endorsed by Mathijs 2026-08-10 — the cocycle reading stays categorically clean and buys cohomology): design questions collected in aips/accepted/0005-dirty-cocycle.md; mechanization targets are Expo 3’s three badge claims. Pricing details stay a Mathijs call (cache-v0 call 2).
- **B3 — a checked complexity claim on ordinary-looking code.** Unchanged horizon deliverable — needs the calculus and a minimal elaborator; B2 and B4 now tell us what the demo should claim.
- **B6 — the strong claim.** Thesis of the monograph; its evidence is B2 (done), B4 (largely done), and the **empirical berry** — **SLOTTED**: Rust microbench of discriminating cases (pointer-chasing vs streaming) against model predictions, plus the dimension measurement: the log-log latency/capacity slope of a real memory hierarchy estimates the machine’s ambient dimension (expected between 2 and 3; shannon-programme §2).

- The **live paper** with its claims ledger remains the basket; the expo series is its front-porch and already carries the receipt mechanism.

Worked-examples ladder

1. **zip** — DONE (mechanized, incl. distance zip + [59]).
2. **FVec push/copy** — DONE (copyWord, doubling copy closed form).
3. **mergesort** — SLOTTED: desk-proved honest cost ($\Theta(N \cdot w \cdot \log N)$ bit-cost) as a badged expo/ledger entry first; Lean later.
4. **transpose** — queued (desk-level layout/boundary-spike study).

Calculus-tier (deferred): splay, union-find, path-copying RB tree, real-time queue.

3. Sequence (current slots, decided 2026-08-10)

1. **Shannon move 1** — block-adaptive re-mounting achieves the entropy rate. Claude designs the statement + lemma ladder, sol executes. Flagship next theorem.
2. **Mergesort ledger entry** — desk-first per the ladder; terra-shaped after a short spec. Independent axis, runs parallel to 1.
3. **Empirical berry** — Rust bench + dimension measurement; sol-shaped after design. Rust earns its place here.
4. Then: C = 1 paragraph bundled into the Expo 2 correction round; Euclidean variants $k = 2, 3$ (+ Campbell/Rényi as their entropy story); dirty-model design $\rightarrow$ mechanization track (Expo 3's badges).

Blocked on Mathijs: cache-v0 call 2 (unlocks phase 3 cliff + dirty pricing details); Expo 1+2 corrections; Letter 2 closing question; veto 5/7.

4. Open questions

1. **Lean-first?** RATIFIED (2026-08-09; spike landed same day). The executable Lean definition doubles as the simulator; Rust enters with the empirical berry (slot 3).
2. **Standalone machine-model paper:** cut early, or keep as monograph chapters? The stated decision point ("after theorems 1–2 are mechanized") is passed — **ripe for Mathijs**.